However, if we dig out our previous example of figuring out the number based on numerous styles of handwritten digits, this task is one of mammoth proportions, so it cannot be accomplished simply employing shallow machine learning. Deep learning and deep neural networks shine in such examples where the computer or software needs to effectively cope with the immense variability of complex features and the even messier nature of interactions between these features.

Why should we know about deep learning

We have reached a point in our technological quests where we want everything to be smart, intuitive and convenient. Nearly every major industry in the world will be affected and try to inculcate Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning in their daily activities, deep learning being at the forefront since it is the next big thing. Fields like healthcare, legal and more are driven towards inculcating deep learning in a massive way.

Automation is the name of the game and deep learning not only automates programmed features but also envelopes uncovered ones since it sports its own intelligence. Deep learning is still in its infancy stages and there is still so much more it can accomplish. Understanding the nuances of deep learning can give you an insight into the future, what matters, what doesn't and what will the future transpire to be.

Additionally, deep learning is beginning to become the jack of all trades, since it is used extensively in every area. It can be applied to any domain which requires data analysis and automated actions to take place based on analysis. Some riveting research domains in deep learning include generative adversarial network, transfer learning, action recognition for sentiment analysis, deep reinforcement learning, image classification, image semantic segmentation and more.

Consumer-based applications

Deep learning sees applications in various tasks and activities that are ingrained in a normal individual's day-to-day life. It is capable of identifying and generating images, generation of speech and language recognition and is even used in concurrence with reinforcement learning that attempts to match human-level performance in games like Go, Dota 2 and Quake III.

Deep learning has transpired to become the foundation of modern online services. For example, it allows Siri, Google Assistant or Alexa to